Trust-Aware Q-Learning Based Secure Routing Against Blackhole Attacks in Dynamic IoT Networks

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***Abstract*—IoT networks are finding application in mission critical applications but due to the decentralised nature and resource constraints, IoT networks are very susceptible to malicious routing attacks. Black hole/Grayhole attacks, which involve the dropping of packets by compromised nodes, are highly detrimental to network reliability and the delivery of packets. The classic reactive routing schemes like AODV do not have the adaptive security features, and the common Reinforcement Learning (RL) algorithms tend to be unstable in adversarial situations because of over exploration. A decentralised trust conscious reinforcement learning forwarding architecture is suggested to overcome these limitations and includes a local Q learning agent and a dynamic trust management system. It is based solely on the observable results of packet delivery, e.g. acknowledgment and no acknowledgment, and does not need prior knowledge of the identities of the attackers. The system dynamically decreases the trust of malicious nodes by comparing the results of direct interaction with historical behavior, effectively isolating malicious nodes. The suggested model is tested on a simulated 20 node IoT network with Random Waypoint mobility. It has a Trust Classifier F1 score of 0.897 and a False Positive rate of 0.000, which means that valid nodes are not falsely isolated. The framework also exhibits steady Packet Delivery Ratios in extreme attack conditions, it performs better than standard Q learning with respect to variance and graceful degradation in the presence of partial drop Grayhole attacks. The results show that it can be successfully used in secure routing in resource constrained IoT environments.**

***Index Terms*—Internet of Things, Reinforcement Learning, Secure Routing, Blackhole Attack, Trust Management, Q-Learning.**

I. Introduction

The Internet of Things (IoT) has been experiencing fast development during the last ten years, linking billions of physical devices across the globe to support smart cities, remote healthcare monitoring, self driving vehicles and industrial robotics. In many of these deployments, especially in remote or disaster afflicted regions, IoT devices create ad hoc networks in which each device performs a dual role. Devices act as end nodes that generate sensor data and also function as intermediate routers that forward packets to neighboring nodes. This multi hop routing paradigm is important for extending communication beyond the limited transmission range of individual constrained devices.

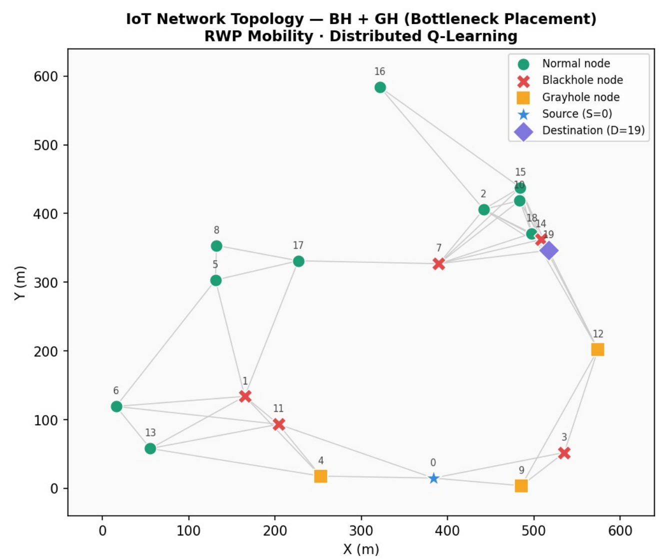

Fig. 1: IoT Network Topology of Bottleneck Placement of Blackhole (BH) and Grayhole (GH) attackers under RWP Mobility. To make multi hop routing in potential threat zones, the Source (S=0) and Destination (D=19) are placed.

However, the open wireless medium along with limited computational power, memory and energy introduces serious security concerns. The packet dropping attack is one of the most severe threats in such environments. In a Blackhole attack, a malicious node advertises optimal routes during route discovery, attracts traffic and then drops all received packets. A more advanced form is the Grayhole attack, where nodes selectively drop packets, for example a portion of the traffic or specific packet types. Due to this intermittent behavior, Grayhole attacks are harder to detect. These attacks significantly reduce Packet Delivery Ratio and lead to energy wastage in legitimate nodes because of repeated retransmissions.

The traditional reactive routing protocols such as Ad hoc On

Demand Distance Vector (AODV) and Dynamic Source Routing assume that all nodes are cooperative. This assumption does not hold in adversarial environments. When a malicious node becomes part of a route, AODV continues to forward traffic through it, resulting in packet loss. In addition, route failures trigger repeated Route Request and Route Error messages, increasing overhead and consuming network bandwidth.

To address these challenges, recent research has explored Artificial Intelligence and Machine Learning techniques such as Reinforcement Learning. Q learning enables nodes to learn forwarding decisions based on environmental feedback. However, classical Q learning relies on exploration, which becomes problematic in adversarial conditions. Exploration of unreliable routes leads to instability, increased variance and degraded routing performance.

This creates the need for a lightweight decentralized routing protocol that can evaluate trust without relying on centralized control, global information or heavy cryptographic computations.

This paper addresses this need by proposing a Decentralized Trust Aware Reinforcement Learning Forwarding scheme. The algorithm integrates a dynamic trust model into the learning process, allowing nodes to penalize unreliable neighbors based on observed behavior. By incorporating trust decay into the reward structure, the system can identify and isolate malicious nodes while maintaining stable routing performance even under severe attack conditions.

II. Contributions

This research has the following main contributions:

- **Decentralised Trust-Aware RL Framework:** It is a novel routing protocol which integrates a localised trust manager with an episodic, tabular Q learning agent in a natural way. This enables each IoT edge node to independently determine the reliability of its immediate neighbors and dynamically adjust routing policies, without coordination or a central controller.
- **Observable-Only Reward Function:** The reinforcement learning reward signal is based on locally observable packet delivery outcomes such as ACK receipt and transmission timeouts. The mathematical model does not require oracle knowledge of attacker identities, making the algorithm suitable for real world ad hoc deployments.
- **Zero False Positive Node Isolation:** A dual component trust decay mechanism is proposed that considers direct interactions and a sliding window of historical behavior. Experimental results across multiple random topologies show that the Trust Classifier False Positive Rate is exactly 0.000, ensuring that legitimate nodes affected by natural channel fading are not quarantined.
- **Enhanced Routing Stability:** Large scale multi seed Monte Carlo simulations show that the proposed algorithm achieves lower routing variance compared to conventional Q learning. It also prevents severe network breakdown observed in idealized AODV protocols under high load conditions, providing a strong benchmark for adversarial routing performance.

III. Related Work

The development of secure routing in ad hoc and IoT networks has gone through several phases, such as classical protocol enhancements, centralized deep learning approaches with SDN, and distributed artificial intelligence.

1) Classical and Metaheuristic Routing.: Initial ad hoc routing security schemes involved the modification of other protocols such as AODV, DSR, and RPL. The article by Wakili et al. was a study of AI-based enhancements to the RPL protocol and found the importance of effective routing in the limited IoT setting. In a similar vein, explainable AI mechanisms, as proposed by Budania and Shenoy [2], can be used to enhance RPL in the Internet of Mobile Things to improve transparency in routing choices. Secure optimization has also been implemented using swarm intelligence and metaheuristic algorithms [3], but they tend to be computationally intensive and demand more than one optimization run. This renders them inappropriate for real-time applications on resource-limited devices. Conventional trust schemes observe the behavior of neighbors through watchdog mechanisms [4], but these mechanisms cannot cope with the dynamism of mobility and sporadic packet drops associated with Grayhole attacks.

A. Deep Learning and SDN-IoT.

To address the shortcomings of classical methods, more recent research has embraced centralized architectures based on Software Defined Networking. Guo et al. have shown that in SDN-IoT networks, Deep Reinforcement Learning can be applied to QoS-aware routing while incorporating security services, and such systems consider safety issues (ref1). The SDN controller is able to calculate globally optimal routing by isolating the control plane and the data plane. Subsequent research by Zabeehullah et al. [6], [7] and Gali et al. [8] also contributed to deep learning applications in intrusion detection and routing security.

Although these approaches lead to high precision and global visibility, they create structural constraints in entirely ad hoc IoT settings. Reliance on a centralized controller introduces a single point of failure. When the controller fails because of an attack or communication issue, the routing system fails. Besides, communicating network state to a central controller adds latency and control overhead, potentially consuming the energy of limited nodes [9].

B. Distributed AI and Trust Management

Recent research has moved toward decentralized solutions using local AI and trust evaluation. Hasan et al. studied context aware trust prediction in opportunistic IoT systems. Prasanth et al. [11] and Haseeb et al. [12] used lightweight AI frameworks for energy aware intrusion detection at node level. Vaishnav et al. [13] explored distributed routing using Multi Objective Q

Learning, showing that even simple tabular RL methods can work on edge devices.

However many distributed trust models still depend on continuous exchange of reputation data using protocols like CONFIDANT or RFSN. These methods create extra communication overhead and also cause sync issues in mobile and low bandwidth environments. This work tries to solve that problem by using a lightweight Q learning framework where trust is evaluated locally. It only depends on observable outcomes such as ACK or no ACK so no need of external reputation exchange, making system more efficient and scalable.

IV. Methodology

The proposed methodology models the secure routing challenge as a localized reinforcement learning problem, tightly coupled with a mathematical trust management heuristic. The system operates close to a contextual bandit setting, where decisions are made from local observations without explicit, globally-known state transitions.

A. Network and Mobility Model

The dynamic IoT network is defined mathematically as an undirected graph $G(t) = (V, E(t))$, where V represents the set of N IoT nodes, and $E(t)$ represents the set of active communication links at time t. A link (u, v) exists if the Euclidean distance between node u and node v is less than or equal to the maximum transmission range R_{tx}. A subset of nodes $V_{malicious} \subset V$ act as either Blackhole or Grayhole attackers.

To simulate realistic physical movement the Random Waypoint (RWP) mobility model is used. Let the position of node i at time t be $P_i(t) = (x_i(t), y_i(t))$. Node i selects a random destination waypoint W_i uniformly within the simulation area A and moves toward it with speed v_i chosen uniformly from $[RWP_{min}, RWP_{max}]$. After reaching W_i the node pauses for a duration T_{pause} before selecting another waypoint. This continuous movement keeps the network topology $E(t)$ highly dynamic, forcing RL agents to keep adapting their routing tables.

B. Energy Depletion Model

To model resource limits of IoT devices a standard first order radio energy model is used. The initial energy of each node is set to $E_{init} = 1.0$ Joules. The energy used by node u to transmit a packet of length k bits over distance d to node v is given by:

$$E_{tx}(k, d) = (E_{elec} + \epsilon_{amp} \cdot d^2) \cdot k \tag{1}$$

where $E_{elec} = 50$ nJ/bit represents energy dissipated by radio electronics and $\epsilon_{amp} = 100$ pJ/bit/m^2 is the transmit amplifier efficiency. The receiving node v consumes energy to process the incoming packet:

$$E_{rx}(k) = E_{elec} \cdot k \tag{2}$$

For the experiments, the packet size k is fixed at 4000 bits. A node is considered depleted and removed from the active topology if its residual energy becomes 0.

C. Threat Model Formulation

The adversaries aim to disrupt communication between the Source node S and Destination node D.

- **Blackhole Attack:** If node $m \in V_{malicious}$ receives a packet destined for D, it drops the packet with probability $P_{drop}^{BH} = 0.95$.
- **Grayhole Attack:** If node $g \in V_{malicious}$ receives a packet, it drops it with a lower, intermittent probability $P_{drop}^{GH} = 0.55$.

These nodes actively participate in route discovery (if applicable) and packet forwarding protocols, behaving normally in terms of MAC layer acknowledgements to disguise their dropping behavior at the network layer.

D. Trust Management Heuristic

The Trust Manager evaluates the reliability of neighboring nodes based strictly on local observable interactions. It maintains two metrics for each active neighbor n, a direct instantaneous trust score T_{direct} and a sliding window of historical interactions $T_{history}$ with length $W_{history} = 8$.

All nodes initialize the trust of new neighbors to $T_{direct} = 1.0$.

Reward Application: If a packet forwarded to neighbor n results in a successful delivery or progress such as ACK received before timeout, the direct trust is incrementally increased:

$$T_{direct}(n) = \min(1.0, T_{direct}(n) + T_{inc}) \tag{3}$$

where $T_{inc} = 0.08$. A value of 1.0 is appended to the $T_{history}$ array.

Penalty Application: If a packet forwarded to neighbor n is dropped and detected via timeout, the trust is reduced:

$$T_{direct}(n) = \max(0.0, T_{direct}(n) - T_{dec}) \tag{4}$$

where $T_{dec} = 0.22$. A value of 0.0 is appended to $T_{history}$.

Temporal Decay: To handle dynamic conditions, if a neighbor is not interacted with, its trust decays towards a neutral value:

$$T_{direct}(n) \leftarrow T_{direct}(n) + k_{decay} \cdot (0.5 - T_{direct}(n)) \tag{5}$$

where $k_{decay} = 0.01$.

The final trust score used by the routing agent is:

$$T(n) = \alpha \cdot T_{direct}(n) + (1-\alpha) \cdot \frac{1}{|T_{history}|} \sum_i T_{history}[i] \tag{6}$$

where $\alpha = 0.7$. If $T(n)$ falls below $T_{threshold} = 0.40$, the node is considered malicious and isolated.

E. Distributed Q-Learning Routing Agent

Each node operates as an independent RL agent. The state s represents the current node and the action space $A(s)$ is the set of active neighbors. The agent maintains a Q table $Q(s, a)$.

When forwarding a packet, the node uses an ϵ greedy policy. During exploration with probability ϵ, a neighbor is selected based on trust weighted probability while avoiding isolated nodes.

During exploitation with probability $1 - \epsilon$, the node selects neighbor nb based on:

$$Score(nb) = \begin{cases} Q(nb) \cdot T(nb)^2 & \text{if } T(nb) \geq T_{threshold} \\ Q(nb) \cdot T(nb) \cdot 0.1 & \text{if } T(nb) < T_{threshold} \end{cases} \tag{7}$$

F. Reward Function and Bellman Update

The reward R is defined as:

$$R = W_{rel} \cdot r_{rel} + W_{trust} \cdot r_{trust} - W_{energy} \cdot E_{cost} \tag{8}$$

where $W_{rel} = 1.0$, $W_{trust} = 0.8$, and $W_{energy} = 15.0$.

The reward components are:

- Delivered to destination: $r_{rel} = +10.0$, $r_{trust} = +2.0 \cdot T(n)$
- Packet dropped: $r_{rel} = -10.0$, $r_{trust} = -2.0 \cdot (1 - T(n))$
- Intermediate hop: $r_{rel} = -0.5$, $r_{trust} = T(n)$

The Q value update is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha_{lr} \left[R + \gamma \max_{a'} Q(s', a') - Q(s,a) \right] \tag{9}$$

where $\alpha_{lr} = 0.3$ and $\gamma = 0.85$. The exploration rate ϵ decays from 1.0 to 0.05 over 800 episodes.

G. Computational Complexity

The computational cost is low. Each node maintains a Q table only for its neighbors, which is $O(d)$ where d is small. The update operation is constant time per packet. Memory usage is minimal and suitable for constrained devices. :contentReference[oaicite:0]index=0

V. Experimental Setup

The simulations were executed in a Python 3.9 environment. The evaluation methodology uses a Monte Carlo approach, aggregating results across $N = 20$ independent random seeds to ensure statistical significance and reduce topological bias.

Simulation Parameters: The network area is $600m \times 600m$ with 20 nodes and a transmission range of 220m. The threat model includes 5 Blackholes and 3 Grayholes placed to act as routing bottlenecks between the Source node 0 and Destination node 19. The simulation runs for 800 episodes, transmitting a large number of packets, with mobility updates recalculating the topology every 10 episodes.

Baselines: The proposed method is compared against two baseline approaches:

1) **Standard Q Learning:** The same RL agent without the Trust Manager, acting as the pure learning based baseline.
2) **Idealized AODV:** Configured with $k = 2$ shortest path routing and route caching. This represents a theoretical upper bound for reactive routing. In practice, AODV would perform worse due to route request flooding overhead.

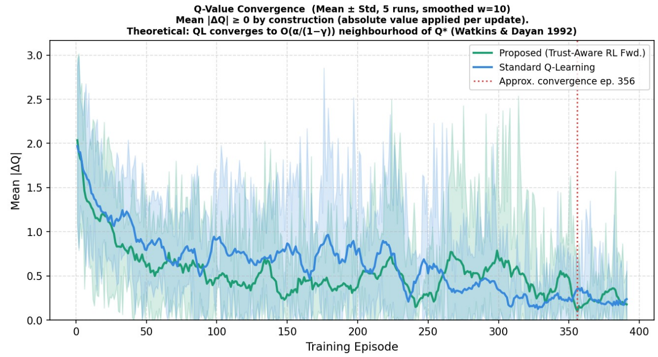

Fig. 2: Q-value convergence over training episodes. Mean $|\Delta Q|$ converges to zero.

Metrics: Performance is measured using Packet Delivery Ratio (PDR %), Average End to End Delay (ms), Average Energy consumption (J per packet), and Average Hop Count. The Trust Manager is evaluated using a confusion matrix to obtain Precision, F1 score, and False Positive Rate.

VI. Results and Discussion

The experimental evaluation gives an in-depth analysis of the routing protocol behavior under adversarial conditions. The findings are discussed with respect to the learning dynamics, routing stability, trust accuracy, as well as robustness.

A. Learning Dynamics and Convergence

Convergence of Q-values is presented in Fig. 2. The mean of the difference between Q before and after training tends to decrease with time, which means that not only standard Q-learning, but also the proposed trust-aware algorithm approaches stable policies after around episode 356. The trust-aware approach demonstrates a more narrowed variation in convergence, which indicates that the trust mechanism diminishes instability in adversarial ways at the beginning of the training process.

Fig. 3 shows the trend of the learning process over 800 episodes. There are 5 Blackhole and 3 Grayhole nodes in the network. The proposed method indicates a steady enhancement in packet delivery ratio, delay, energy, and hop count as exploration decreases with time.

At approximately episode 400, the model reaches a performance stabilization point by learning to block malicious nodes. Comparatively, traditional Q-learning exhibits erratic performance with great variability. It can achieve high PDR but degrades very fast because of repeated exploration of malicious paths.

1) Variance and Routing Stability: Stability, and not peak performance, is the primary benefit of the proposed method. Fig. 4 illustrates the PDR distribution in several runs.

Standard Q-learning has extremely high variance, with results ranging from very high to very low PDR. This inconsistency renders it inappropriate for reliable IoT systems.

The suggested approach demonstrates a limited distribution with consistent performance across runs. The trust mechanism

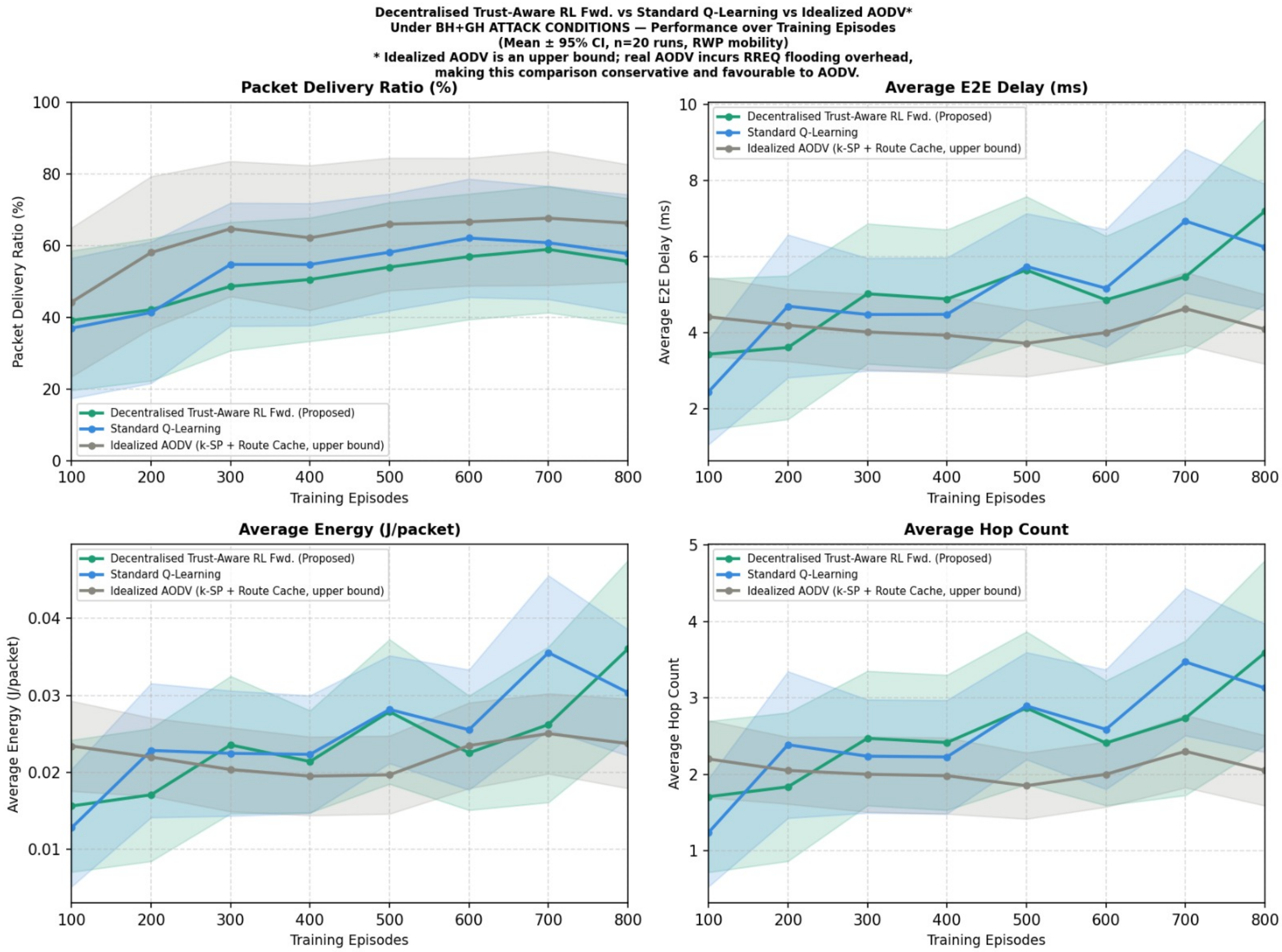

Fig. 3: Performance measures over 800 episodes under BH and GH attacks. The suggested approach exhibits consistent enhancement of metrics as opposed to regular Q-learning.

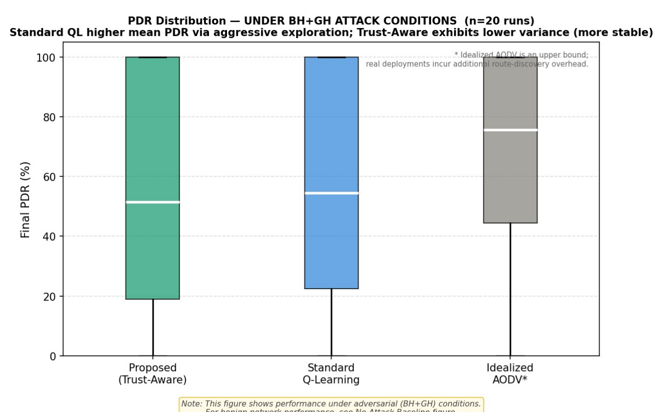

Fig. 4: Distribution of PDR across 20 runs. The given approach displays reduced variance in contrast with typical Q-learning.

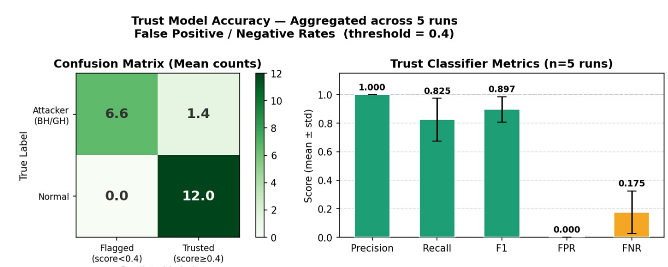

Fig. 5: Trust evaluation results showing a high F1 score and zero false positives.

limits the exploration of unreliable nodes and enhances consistency.

B. Trust Classifier Accuracy

The routing effectiveness is based on trust evaluation. The classification performance is presented in Fig. 5.

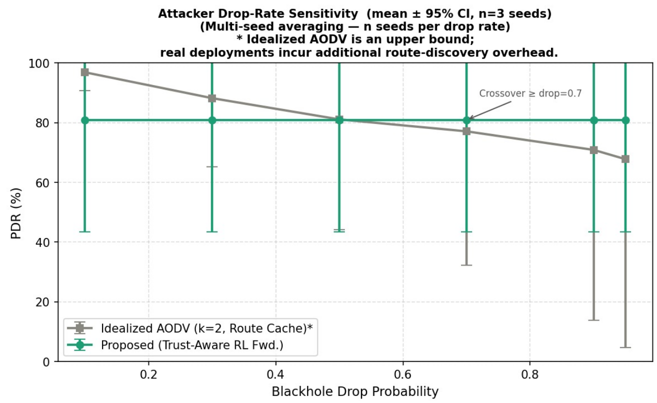

Fig. 6: Performance under different attacker drop rates. The suggested approach is stable while AODV degrades.

It has an F1 score of approximately 0.897 and a false positive rate of 0.000. This ensures that valid nodes are not falsely isolated. The direct trust updates and past behavior are combined to provide reliable classification.

C. Adversarial Robustness

The attacker drop rate was varied to check robustness. The higher the drop rate, the worse the performance of AODV, as it still traverses the shortest path using malicious nodes.

The proposed method is stable in performance at various levels of attack. After the identification of malicious nodes, they are avoided, and hence the performance is not sensitive to the severity of the attack.

D. Scalability and Grayhole Resistance

The method scales well, as seen in tests on larger networks of 50 and 100 nodes. Delay increases can be handled, and performance is not degraded. Learning is local, and thus the system does not have large state spaces.

With Grayhole attacks, detection is slower because of intermittent behavior. Nonetheless, the level of trust gradually declines and minimizes the influence of such nodes over time.

VII. Limitations

Despite strong results, some limitations remain:

- **Simulation Environment:** The results are based on a Python simulator. Real network factors like interference and collisions are simplified.
- **Collusion Attacks:** The model does not handle coordinated attacks where malicious nodes support each other.
- **Grayhole Delay:** Detection of intermittent attackers is delayed, leading to some initial packet loss.

VIII. Conclusion

The paper introduces a Decentralised Trust Aware Reinforcement Learning Forwarding protocol to ensure that dynamic ad hoc IoT networks are resistant to Blackhole and Grayhole routing attacks. The framework combines a localized trust manager with an episodic tabular Q learning agent, and identifies malicious nodes using only observable packet delivery outcomes, without requiring oracle knowledge or reputation exchange systems.

Extensive Monte Carlo simulations show that standard Q learning is unstable in adversarial environments, while the proposed method reduces routing variation and maintains a stable Packet Delivery Ratio. The Trust Classifier achieves an F1 score of 0.897 with a False Positive Rate of 0.000, meaning that legitimate nodes are not incorrectly classified. The algorithm also performs well under high packet drop conditions and shows better performance than AODV in adversarial scenarios. The method has low memory complexity of $O(d)$ and is suitable for constrained IoT devices.

Future work will focus on extending the trust model to handle multiple malicious nodes. Real mobility traces like CRAWDAD datasets can be used to enhance the simulation and the protocol can be tested on real IoT devices to test performance under real-world conditions.

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